

MSDS OF 2,4-D DIMETHYL AMMONIUM

84% W/V SL

1. PRODUCT AND COMPANY IDENTIFICATION

SUPPLIER: CHANGZHOU WINTAFONE CHEMICAL CO., LTD.

HERBICIDE

Use A selective water soluble concentrate hormone type herbicide for the post-emergence control of broadleaf weeds in crops and grass pastures as indicated on the label.

2. COMPOSITION / INFORMATION ON

INGREDIENTS

Active ingredient 2,4-D amine

Chemical Name 2,4-D-dimethylammonium

(IUPAC)

CAS No. 2008-39-1

Chemical Family Aryloxyalkanoic acid

Chemical Formula $C_{10}H_{13}Cl_2NO_3$ (Mol. wt.: 266.1)

UN no. 3082

IMDG CLASS : 9

EEC classification Xn , Xi

R phrases R 22, R 36/37

3. HAZARD IDENTIFICATION

Toxicity class:

WHO (a.i.) II; EPA (formulation) II

ADI (JMPR): 0.01 mg/kg b.w.

NOEL: 5 mg/kg b.w. for rats and mice (2 y)

Main hazard Eye irritant.

Biological hazards:

Harmful to algae.

Eye contact:

Very irritating to eyes.

Skin contact:

Not irritating to skin.

Ingestion:

Harmful if large amounts are swallowed.

Inhalation:

Moderately irritating to respiratory tract.

4. FIRST AID MEASURES

Inhalation:

In case of inhalation, remove source of contamination, or leave contaminated area to fresh air as rapidly as possible.

Keep victim from contact for at least 2-3 days.

Skin contact:

If irritation occurs, remove contaminated clothing, shoes and leather goods (e.g. watchbands, belts). Gently wipe off excess chemical. Wash skin gently and thoroughly with water and non-abrasive soap. Seek medical advice immediately if irritation persists.

Eye contact:

Immediately flush eyes with gently flowing lukewarm water or saline solution for 15 minutes, holding the eyelids open. Seek medical attention.

Ingestion:

Unlikely to occur under occupational conditions. In case of deliberate ingestion, have victim rinse mouth thoroughly with water. Do not induce vomiting. Give plenty of water to drink.

Seek medical advice immediately. If breathing has stopped, apply artificial respiration.

Advice to the physician

No specific antidote.

Supportive care. Treatment based on judgement of physician in response to symptoms of patient. If lavage is performed, suggest endotracheal and/or oesophageal control.

Danger from lung aspiration must be weighed against toxicity when considering emptying the stomach.

5. FIRE FIGHTING MEASURES

Extinguishing media:

Small fires: Carbon dioxide, dry chemical powders, regular foam and water spray.

Large fires: Water spray, fog or regular foam. Move containers from fire area if you can do it without risk.

Special hazards:

The material does not burn or burns with difficulty. It is not explosive. Should the

chemical be involved in a general fire, ensure chemical protective clothing are used. It can produce toxic fumes of hydrogen chloride, which forms mists of hydrochloric acid with moisture and phosgene; and carbon monoxide.

Protective clothing:

Wear suitable personal protective equipment including approved respiratory protection.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions:

Chemical protective clothing usage is advised, i.e. wear neoprene gloves, cotton overalls and safety goggles.

Environmental precautions:

Do not allow spill to contaminate water supplies. Dike far ahead of liquid spills for later disposal.

Large spills:

Keep spectators away. Isolate hazard area and deny entry.

Stay upwind, out of low-lying areas, and ventilate closed spaces before entering.

Cover spill with absorbent material.

Sweep into disposal container. Wash area with detergent and water and follow with clean water rinse. Do not allow spill to contaminate water supplies. Prevent entry into waterways, sewers, basements or confined areas.

Small spills:

Take up with sand or other noncombustible absorbent material and place into containers for later disposal.

Spill/Leak Procedures:

Notify safety personnel, isolate and ventilate area, deny entry, and stay upwind. Shut off all ignition sources.

7. HANDLING AND STORAGE

Handling:

Do not eat, drink, smoke or go to the toilet with pesticidecontaminated hands. Always wash hands thoroughly after handling pesticides or waste.

Storage:

Store in a dry, cool, well-ventilated warehouse in well-labeled containers. Not to be stored next to foodstuffs and water supplies. Keep away from children and animals.

Local regulations should be complied with.

8. EXPOSURE CONTROL / PERSONAL

PROTECTION

Occupational exposure limits:

2, 4-D dimethylamine salt: None established; ACGIH TLV and OSHA PEL are 10 mg/M3 for the acid.

PELs are in accord with those recommended by OSHA, as in the 1989 revisions of the PELs.

Engineering control measures:

It is essential to provide adequate ventilation. The measures appropriate for a particular worksite depend on how this material is used and on the extent of exposure.

Ensure that control systems are properly designed and maintained.

Comply with occupational safety, environmental, fire, and other applicable regulations.

If engineering controls and work practices are not effective in controlling exposure to this material, then wear suitable personal protective equipment including approved respiratory protection.

Respirator:

An approved respirator suitable for protection from dusts and mists of pesticides is adequate. Limitations of respirator use specified by the approving agency and the manufacturer must be observed.

Clothing:

Employee must wear appropriate protective (impervious) clothing and equipment to prevent repeated or prolonged skin contact with this substance.

Gloves:

Employee must wear appropriate synthetic protective gloves to prevent contact with this substance.

Eye protection:

The use of safety goggles is recommended

9. PHYSICAL AND CHEMICAL PROPERTIES

Data as for 840 SL formulation.

Appearance:

Light yellow liquid.

Odour: -

Mild phenolic odour.

Explosive properties:

Not explosive.

Flash point:

Not applicable, water based formulation. UNRESTRICTED -

Relative density: 1,16 g/ml at 20 °C.

10. STABILITY AND REACTIVITY

Storage stability:

2, 4-D is stable at elevated temperatures and at low temperatures. Stable in aqueous solutions (after 18 hours at 20 °C).

Incompatibility:

Oxidizing agents.

Hazardous decomposition products:

Hydrogen chloride, carbon monoxide, carbon dioxide.

11. TOXICOLOGICAL INFORMATION

Acute oral LD50 :

600, 5 mg/kg in rats.

Acute dermal LD50: > 4 347, 73 mg/kg in male rats.

Acute inhalation LC50(4h) : 1,60 mg/l air for 600 SL formulation.

Acute skin irritation:

This formulation did not provoke dermal irritation or corrosion.

Acute eye irritation:

Severe eye and surrounding tissue damage.

Dermal sensitisation:

Probability of sensitization is very low.

12. ECOLOGICAL INFORMATION

ECOTOXICOLOGY :

Daphnia: Practically non-toxic.

The 48 hour LC50 is > 130, 3 mg/ Daphnia magna)

Fish: Practically non-toxic. 96-hour LC50 is >114 mg/

Following data as for 2,4-D amine 600 SL formulation:

Birds: Practically non-toxic.

LD50: 625 to 2 000 mg/kg (bobwhite quail.)

Freshwater algae:

EC50 > 100 mg/ growth rate and biomass.)

Bees:

Oral: 72-hour LD50 >100 µg formulation/bee

NOEC: 25 µg formulation/bee.

Contact : 72-hour LD50 >200 µg formulation/bee

NOEC: 200 µg formulation/bee.

Soil microorganisms:

Low risk to soil microorganisms.

13. DISPOSAL CONSIDERATIONS

Controlled incineration:

2, 4-D is stable under normal temperatures and pressures.

Package product wastes:

Combustible containers should be disposed of in pesticide incinerators or in specified landfill sites. Non-combustible containers must be triple rinsed using the normal diluent at a volume equal to approximately 10% of the drum's capacity.

Add the rinsing mixture to the spray mixture or use the recommended disposal methods. Containers must be punctured and disposed of in specified landfill areas.

14. TRANSPORT INFORMATION

UN NUMBER 3082

IMDG/IMO Class: 9

15. REGULATORY INFORMATION

Symbol : Xn , Xi

Risk phrases :

R 22 Harmful if swallowed.

R 36/37 Irritating to eyes and respiratory system.

Safety phrases:

S2 Keep out of reach of children.

S13 Keep away from food, drink and animal feeding stuffs

S20/21 When using do not eat, drink or smoke.

S36/37/39 Wear suitable protective clothing, gloves and eye/face protection.

16. OTHER INFORMATION

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct.